

Benchmark de Modelos GPT para Detecção de Spam

Dataset

SMS Spam Collection

→ Conjunto público de mensagens SMS rotuladas, criado para pesquisas sobre detecção de spam em dispositivos móveis.

→ Contém **5.574 mensagens** classificadas como **ham** ou **spam**.

→ As mensagens foram coletadas de várias fontes públicas, incluindo o fórum **Grumbletext** (Reino Unido), o **NUS SMS Corpus** (Universidade Nacional de Singapura), a **tese de doutorado de Caroline Tagg**, e o **SMS Spam Corpus v.0.1 Big**.

Almeida, T. & Hidalgo, J. (2011). *SMS Spam Collection*. UCI Machine Learning Repository.

Dataset

SMS Spam Collection

```
id,label,text
1,ham,"Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat..."
2,ham,Ok lar... Joking wif u oni...
3,spam,Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
4,ham,U dun say so early hor... U c already then say...
5,ham,"Nah I don't think he goes to usf, he lives around here though"
6,spam,"FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, £1.50 to rcv"
7,ham,Even my brother is not like to speak with me. They treat me like aids patient.
8,ham,As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
9,spam,WINNER!! As a valued network customer you have been selected to receive a £900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
10,spam,Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
```



UCI Machine Learning Repository

Discover datasets around the world!

 ics.uci.edu

Paper 1

Towards SMS Spam Filtering: Results under a New Dataset

de Tiago A. Almeida, José María Gómez Hidalgo e Tiago P. Silva
UFSCar (Brasil) e Optenet (Espanha), 2013

→ Criação e análise de um novo dataset público e real de SMS (*SMS Spam Collection*) para pesquisas em detecção de spam.

→ Algoritmos testados:

- Naïve Bayes (várias variantes)
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Trees (C4.5, Boosted C4.5)
- Random Forest
- K-Nearest Neighbors (K=1,3,5)
- Multilayer Perceptron (MLP)
- Sequential Minimal Optimization (SMO)
- Expectation-Maximization (EM)
- Boosted Naïve Bayes, PART, MDL

(PDF) Towards SMS Spam Filtering: Results under a New Dataset

PDF | The growth of mobile phone users has lead to a dramatic increasing of SMS spam messages. Recent reports clearly...

[researchgate.net](https://www.researchgate.net/publication/260201100)

Paper 2

SMS Spam Detection using Machine Learning Approach

de Houshmand Shirani-Mehr

Stanford University, 2013

→ Aplicar e comparar diferentes algoritmos de aprendizado de máquina para detecção de spam em mensagens SMS, utilizando o dataset SMS Spam Collection (UCI).

→ Algoritmos testados:

- Naïve Bayes (Multinomial e Bayesiano)
- Support Vector Machine (SVM) — vários kernels (linear, polinomial, RBF, sigmoide)
- K-Nearest Neighbors (KNN)
- Random Forests
- AdaBoost (com Decision Trees)

<https://cs229.stanford.edu/proj2013/ShiraniMehr-SMSSpamDetectionUsingMachineLearningApproach.pdf>

Paper 3

Optimized SMS Spam Detection Using SVM-DistilBERT and Voting Classifier

de Sinar Nadhif Ilyasa, Alaa Omar Khadidos
King Abdul Aziz University (Arábia Saudita), 2024

→ Aprimorar a detecção de spam em SMS combinando SVM com DistilBERT embeddings e um Voting Classifier, avaliando também o impacto da lematização e aumento de dados por sinônimos.

→ Modelos comparados:

- SVM-TF / SVM-TF-Bi / SVM-TF-Bi-Ada / SVM-TF-Bi-XGB / SVM-TF-Bi-Vote
- Proposto: SVM + DistilBERT + Voting Classifier
- Ferramentas adicionais: Optuna (otimização de hiperparâmetros), LIME (explicabilidade)

Optimized SMS Spam Detection Using SVM-DistilBERT and Voting Classifier: A Comparative Study on the Impa...

The rapid growth of digital communication has led to a surge in spam messages, particularly through Short Message Service (SMS). These unsolicited messages pose risks such as phishing and malware,...

Benchmark de Modelos GPT para Detecção de Spam

Objetivo

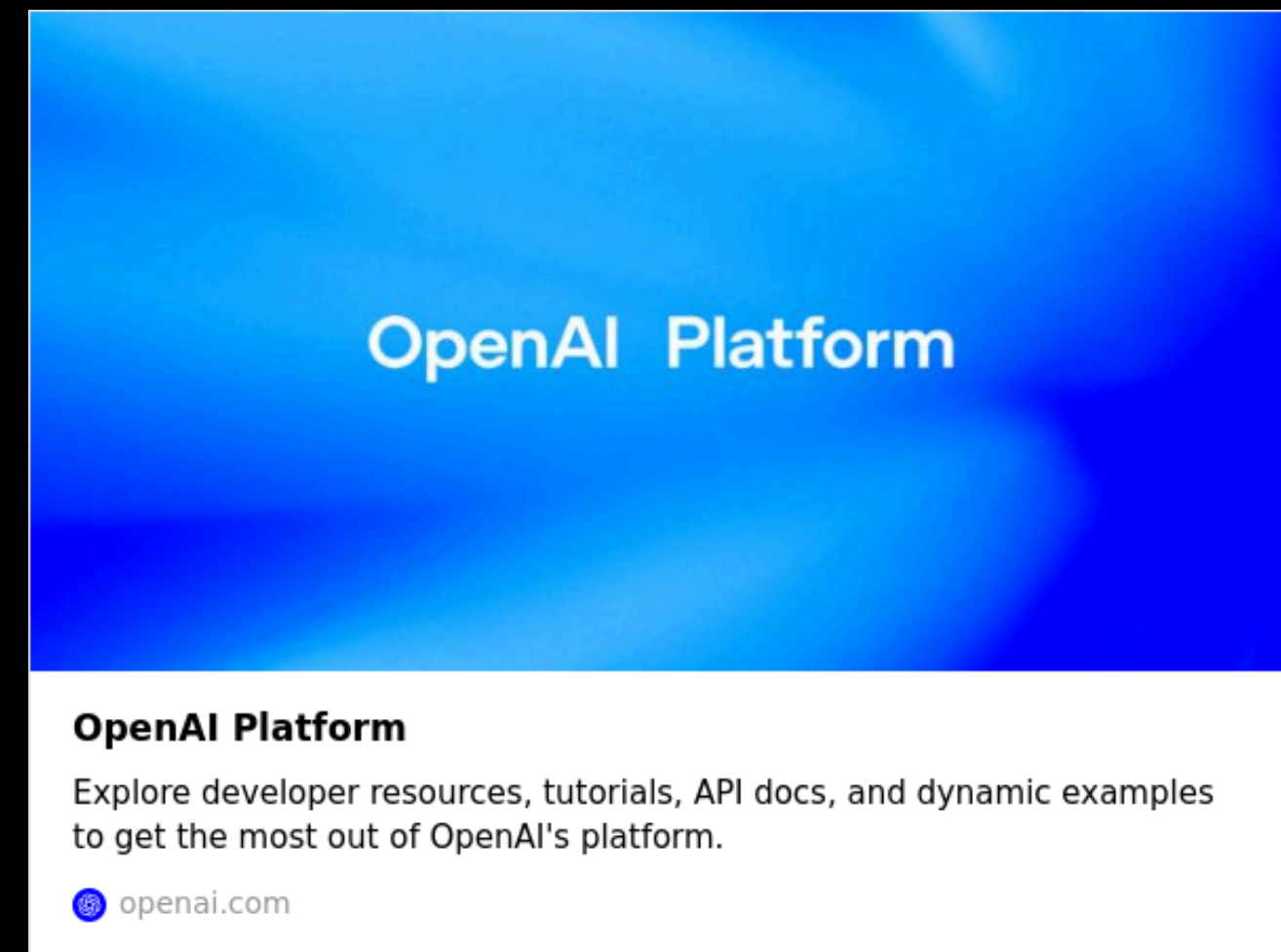
- Realizar um benchmark comparativo de **modelos GPT** na tarefa de **detecção de spam** em mensagens SMS, utilizando o dataset público **SMS Spam Collection** v.1 (UCI).
- Comparar o desempenho entre diferentes versões e tamanhos dos modelos GPT na tarefa de classificação binária (ham/spam).
 - Confrontar os resultados obtidos com abordagens de outras pesquisas.
 - Investigar a explicabilidade dos erros de classificação.

System Prompt

```
SYSTEM_PROMPT = (  
    "You are a binary text classifier for SMS messages. "  
    "Classify each message as exactly one of the following labels:\n"  
    "- ham: legitimate, personal, or non-promotional content\n"  
    "- spam: promotional, fraudulent, or unsolicited content\n\n"  
    "Respond with only the label – 'ham' or 'spam' – without explanation or punctuation."  
)
```


Modelos Utilizados

- gpt-5
- gpt-5-mini
- gpt-5-nano
- gpt-4.1
- gpt-4.1-mini
- gpt-4o
- gpt-4o-mini
- gpt-3.5-turbo



Batches

→ O benchmark utilizou a API de Batches da OpenAI, que permite enviar milhares de requisições de inferência de forma assíncrona e eficiente.

→ Cada requisição é tratada como uma tarefa independente dentro de um arquivo .jsonl

→ Essa abordagem reduz o tempo total de execução e o custo por requisição.

→ O processamento em lotes é essencial para comparar múltiplos modelos LLM sob as mesmas condições, garantindo reprodutibilidade e rastreabilidade.

Batches

batch_6902260ddd048190a2b3d922ef9f2c1c	10/29/2025, 11:34 AM
batch_69022608abec8190b178e7b483cd2c1b	10/29/2025, 11:34 AM
Cancelled	
batch_6902260317908190bdd7f9c11fea5e2b	10/29/2025, 11:34 AM
Failed	
batch_690225fdc7008190b6bed82324c5a07f	10/29/2025, 11:34 AM
batch_690225f8303081909b813aad8ee042af	10/29/2025, 11:34 AM
Cancelled	
batch_690225f2fbc081908565841b0e13bc98	10/29/2025, 11:34 AM
Failed	
batch_690225ed4eb481908126f0b3b4f8950e	10/29/2025, 11:34 AM
batch_690225e7a9448190b75ab4fd29c2a001	10/29/2025, 11:34 AM
batch_690225e16de88190a85e18651a0f068a	10/29/2025, 11:34 AM
batch_690225db5f948190a1309a59cc15746d	10/29/2025, 11:34 AM
Failed	
batch_69021ee1dce0819081a39f5dd7db7803	10/29/2025, 11:04 AM
batch_6902143de2ec81909a9ee3dbebb38446	10/29/2025, 10:18 AM

BATCH

batch_690225fdc7008190b6bed82324c5a07f

Status

Completed

Created at

Oct 29, 2025, 11:34 AM

Endpoint

/v1/chat/completions

Completion window

24h

Completion time

2 hours, 19 minutes

Request counts

5160 completed, 0 failed of 5160 total requests

Files

Input

batch_input_gpt-5-nano.jsonl

Output

batch_690225fdc7008190b6bed82324c5a07f_output.jsonl

Error

WEDNESDAY, OCTOBER 29, 2025

11:34:37

Batch created

11:35:40

Batch in progress

13:45:53

Batch finalizing

13:54:05

Batch completed

Batches

Input / Request

```
{
  "custom_id": "2388",
  "method": "POST",
  "url": "/v1/chat/completions",
  "body": {
    "model": "gpt-5",
    "messages": [
      {"role": "system", "content": "<SYSTEM_PROMPT>"},
      {"role": "user", "content": "<SMS_TEXT>"}
    ]
  }
}
```

```
SYSTEM_PROMPT = (
  "You are a binary text classifier for SMS messages. "
  "Classify each message as exactly one of the following labels:\n"
  "- ham: legitimate, personal, or non-promotional content\n"
  "- spam: promotional, fraudulent, or unsolicited content\n\n"
  "Respond with only the label – 'ham' or 'spam' – without explanation or punctuation."
)
```

```
id,label,text
1,ham,"Go until jurong point, crazy.. Available only in bugis n great wor
2,ham,Ok lar... Joking wif u oni...
3,spam,Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005
T&C's apply 08452810075over18's
4,ham,U dun say so early hor... U c already then say...
5,ham,"Nah I don't think he goes to usf, he lives around here though"
6,spam,"FreeMsg Hey there darling it's been 3 week's now and no word back
chgs to send, £1.50 to rcv"
7,ham,Even my brother is not like to speak with me. They treat me like ai
8,ham,As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettai
*9 to copy your friends Callertune
9,spam,WINNER!! As a valued network customer you have been selected to re
code KL341. Valid 12 hours only.
10,spam,Had your mobile 11 months or more? U R entitled to Update to the
Mobile Update Co FREE on 08002986030
11,ham,"I'm gonna be home soon and i don't want to talk about this stuff
12,spam,"SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and
Reply HL 4 info"
13,spam,"URGENT! You have won a 1 week FREE membership in our £100,000 Pr
net LCCLTD POBOX 4403LDNW1A7RW18"
14,ham,I've been searching for the right words to thank you for this breac
fulfil my promise. You have been wonderful and a blessing at all times.
15,ham,I HAVE A DATE ON SUNDAY WITH WILL!!
16,spam,"XXXMobileMovieClub: To use your credit, click the WAP link in the
xxxmobilemovieclub.com?n=QJKGIGHJJGCBL"
17,ham,Oh k ... i'm watching here:)
18,ham,Eh u remember how 2 spell his name... Yes i did. He v naughty make
19,ham,Fine if that's the way u feel. That's the way its gota b
20,spam,"England v Macedonia - dont miss the goals/team news. Txt ur natio
4txt/ú1.20 POBOXox36504W45WQ 16+"

```


Batches

Output / Response

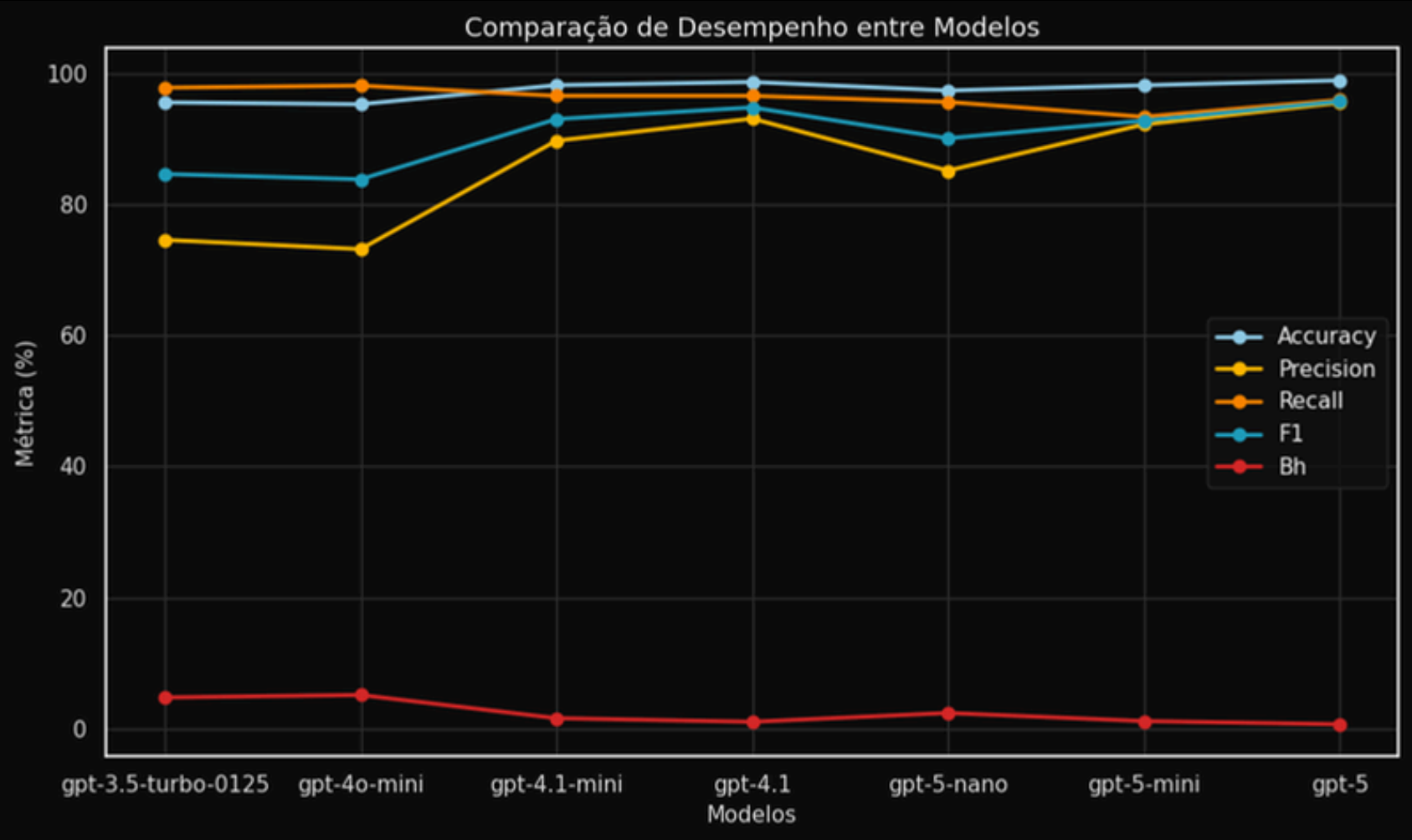
```
{
  "id": "batch_req_XXXXXXX",
  "custom_id": "2388",
  "response": {
    "status_code": 200,
    "request_id": "XXXXXXXX",
    "body": {
      "id": "chatcmpl-XXXX",
      "object": "chat.completion",
      "created": 1761868812,
      "model": "gpt-5-2025-08-07",
      "choices": [
        {
          "index": 0,
          "message": {"role": "assistant", "content": "ham"}
        }
      ],
      "usage": {...}
    }
  },
  "error": null
}
```

```
{
  "id": "batch_req_6903fde5a2208190add6e1312ccad518",
  "custom_id": "2388",
  "response": {
    "status_code": 200,
    "request_id": "006045f79740edafca058ceb9f515203",
    "body": {
      "id": "chatcmpl-CWWqyJPB9M1ujgAkmoydVn1pGIQw",
      "object": "chat.completion",
      "created": 1761868812,
      "model": "gpt-5-2025-08-07",
      "choices": [
        {
          "index": 0,
          "message": {
            "role": "assistant",
            "content": "ham",
            "refusal": null,
            "annotations": []
          },
          "finish_reason": "stop"
        }
      ],
      "usage": {
        "prompt_tokens": 88,
        "completion_tokens": 74,
        "total_tokens": 162,
        "prompt_tokens_details": {
          "cached_tokens": 0,
          "audio_tokens": 0
        },
        "completion_tokens_details": {
          "reasoning_tokens": 64,
          "audio_tokens": 0
        },
        "accepted_prediction_tokens": 0,
        "rejected_prediction_tokens": 0
      },
      "service_tier": "default",
      "system_fingerprint": null
    },
    "error": null
  }
}
```

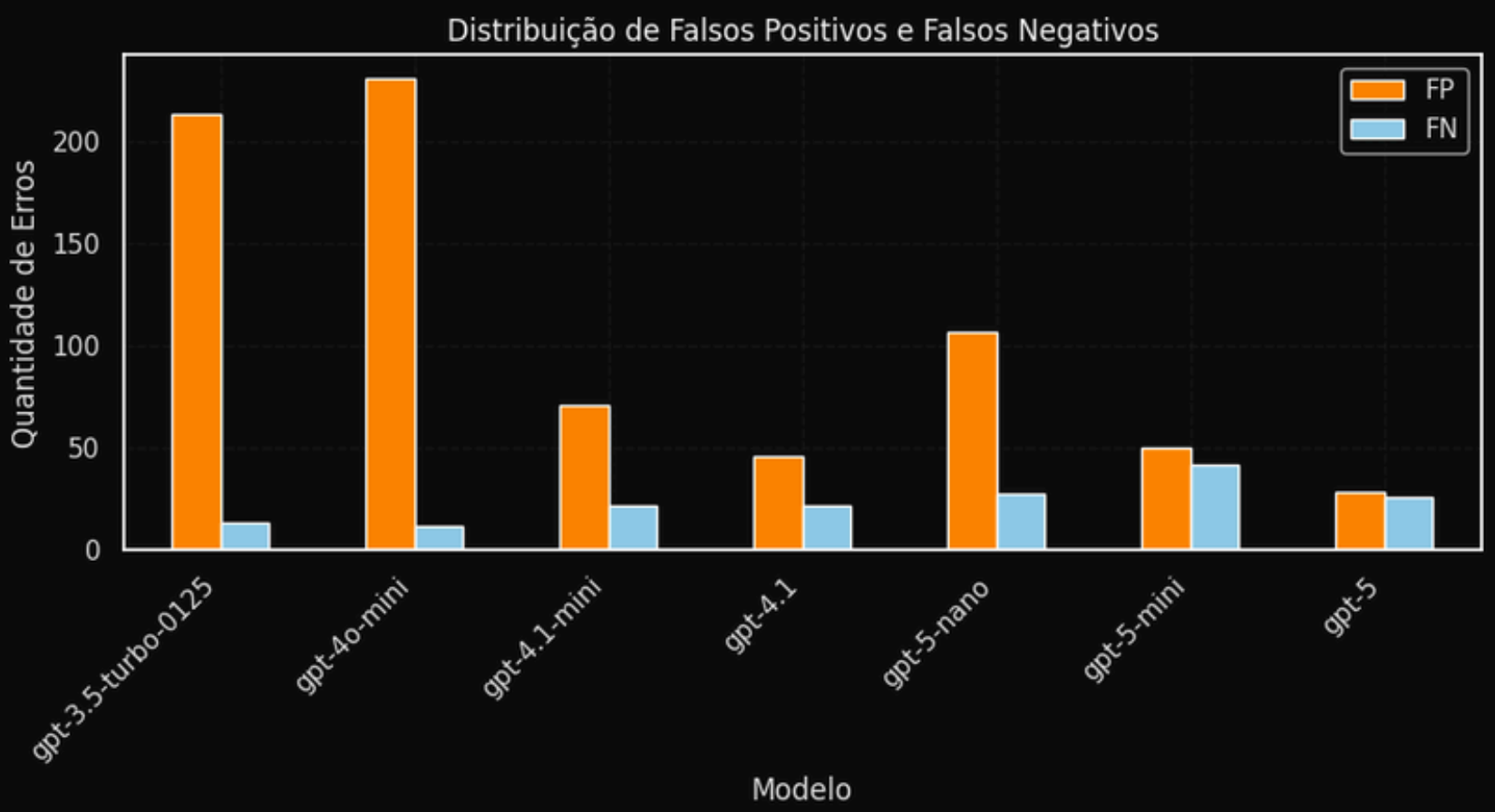
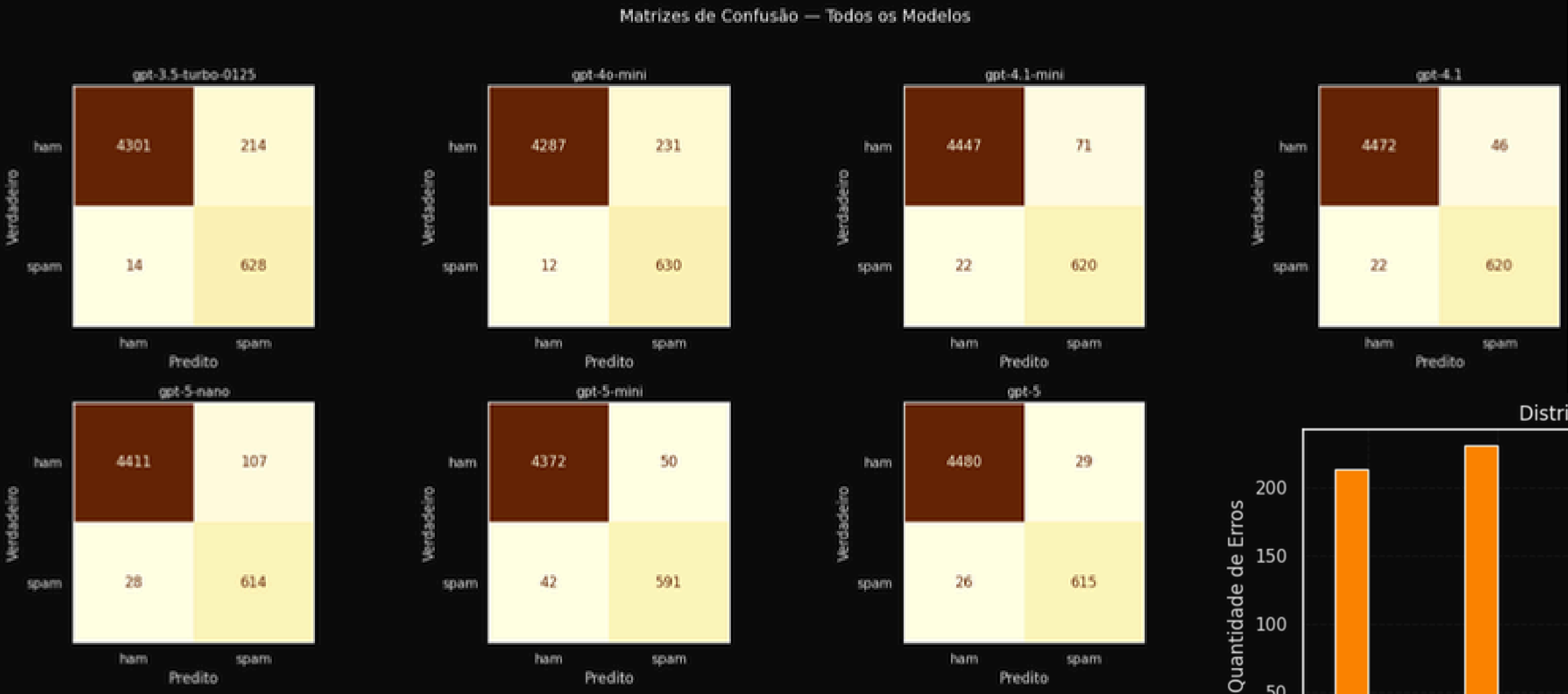
id	prediction
2388	ham
3526	ham
655	ham
1085	ham
2517	ham
2499	ham
3605	ham
1540	ham
2471	ham
3265	ham
808	ham
799	ham
66	spam
2101	ham
2043	ham
4222	ham
1891	ham
3817	ham
3991	ham
2592	ham
4828	spam
4685	ham
466	ham
2440	ham
3312	ham
2313	ham
3132	ham
2655	ham
4031	ham
743	ham
801	ham
654	ham
2858	spam

Resultados

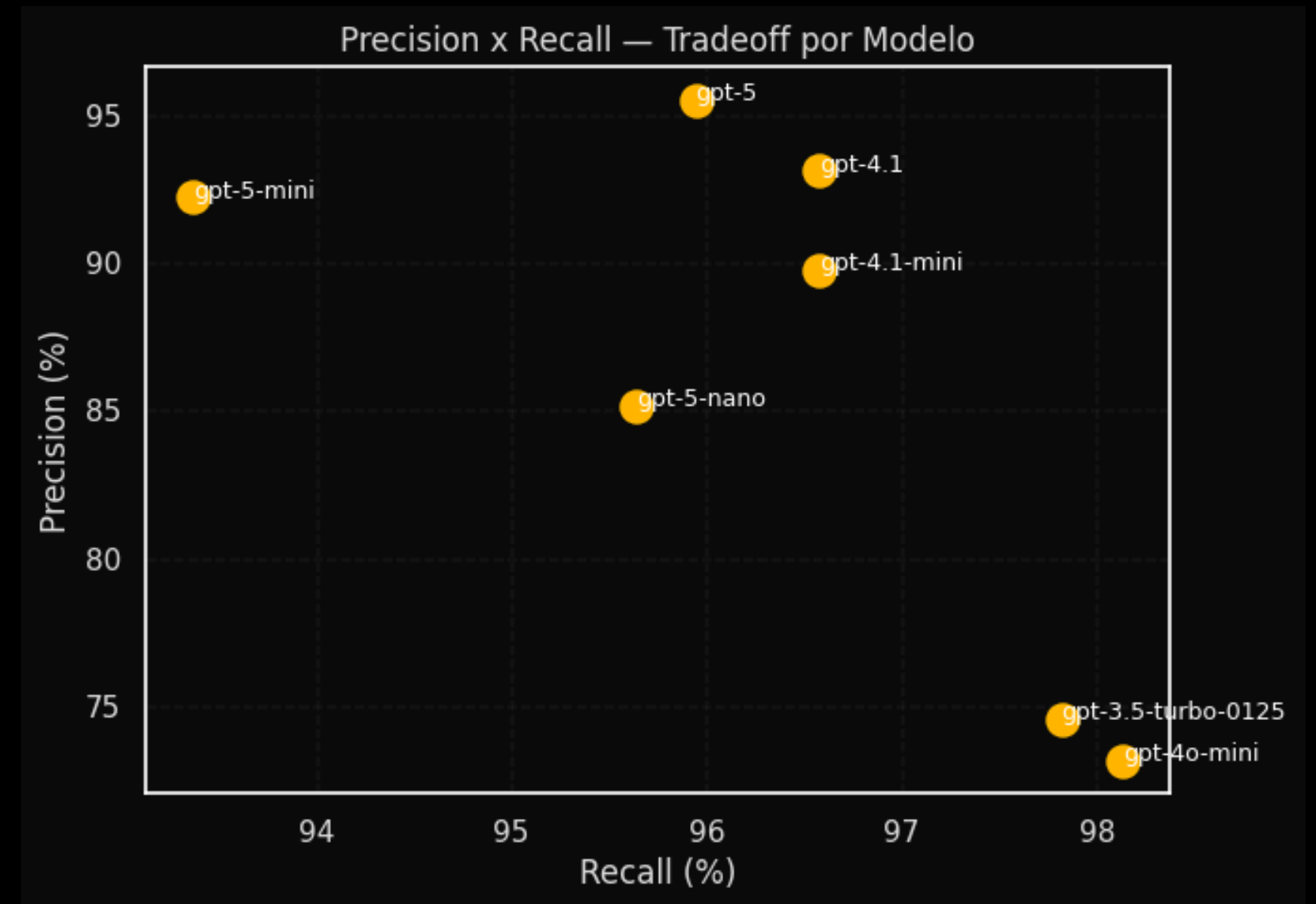
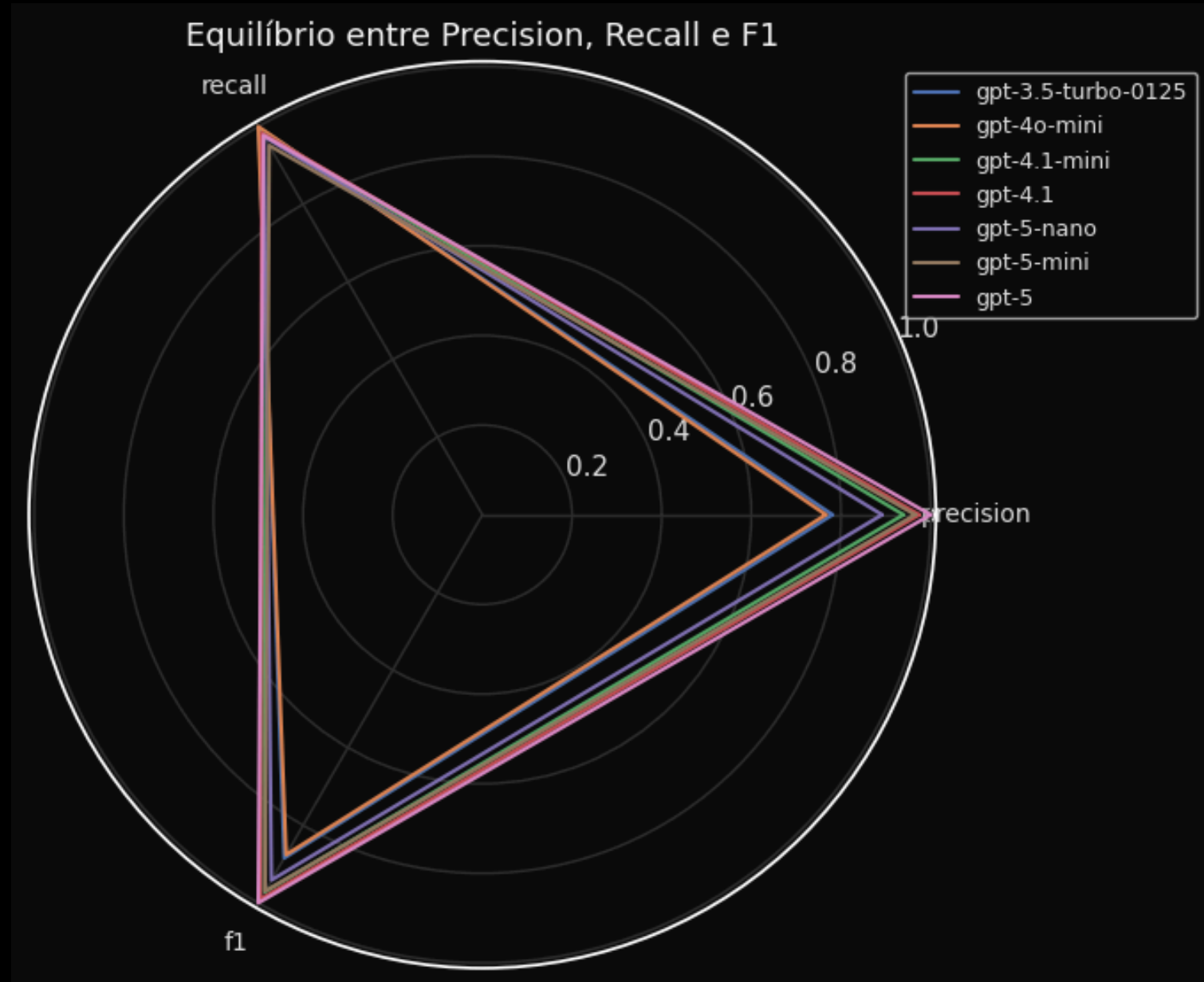
	model	samples	accuracy	precision	recall	f1	bh	mcc
3	gpt-5	5150	98.93	95.50	95.94	95.72	0.64	95.11
6	gpt-4.1	5160	98.68	93.09	96.57	94.80	1.02	94.07
0	gpt-4.1-mini	5160	98.20	89.73	96.57	93.02	1.57	92.07
4	gpt-5-mini	5055	98.18	92.20	93.36	92.78	1.13	91.74
2	gpt-5-nano	5160	97.38	85.16	95.64	90.10	2.37	88.79
1	gpt-3.5-turbo-0125	5157	95.58	74.58	97.82	84.64	4.74	83.14
5	gpt-4o-mini	5160	95.29	73.17	98.13	83.83	5.11	82.34



Resultados



Resultados



Comparações

Paper 1

Classifier	<i>SC</i> %	<i>BH</i> %	<i>Acc</i> %	<i>MCC</i>
Logistic Reg. + tok2	95.48	2.09	97.59	0.899
SVM + tok1	83.10	0.18	97.64	0.893
Boosted NB + tok2	84.48	0.53	97.50	0.887
SMO + tok2	82.91	0.29	97.50	0.887
Boosted C4.5 + tok2	81.53	0.62	97.05	0.865
MDL + tok1	75.44	0.35	96.26	0.826
PART + tok2	78.00	1.45	95.87	0.810
Random Forest + tok2	65.23	0.12	95.36	0.782
C4.5 + tok2	75.25	2.03	95.00	0.770
Bern NB + tok1	54.03	0.00	94.00	0.711
MN TF NB + tok1	52.06	0.00	93.74	0.697
MN Bool NB + tok1	51.87	0.00	93.72	0.695
INN + tok2	43.81	0.00	92.70	0.636
Basic NB + tok1	48.53	1.42	92.05	0.600
Gauss NB + tok1	47.54	1.39	91.95	0.594
Flex NB + tok1	47.35	2.77	90.72	0.536
Boolean NB + tok1	98.04	26.01	77.13	0.507
3NN + tok2	23.77	0.00	90.10	0.462
EM + tok2	17.09	4.18	85.54	0.185
TR	0.00	0.00	86.95	–

Paper 2

Model	SC %	BH %	Accuracy %
Multinomial NB	94.47	0.51	98.88
SVM	92.99	0.31	98.86
<i>k</i> -nearest neighbor	82.60	0.40	97.47
Random Forests	90.62	0.29	98.57
Adaboost with decision trees	92.17	0.51	98.59

Esta Pesquisa

	model	samples	accuracy	precision	recall	f1	bh	mcc
3	gpt-5	5150	98.93	95.50	95.94	95.72	0.64	95.11
6	gpt-4.1	5160	98.68	93.09	96.57	94.80	1.02	94.07
0	gpt-4.1-mini	5160	98.20	89.73	96.57	93.02	1.57	92.07
4	gpt-5-mini	5055	98.18	92.20	93.36	92.78	1.13	91.74
2	gpt-5-nano	5160	97.38	85.16	95.64	90.10	2.37	88.79
1	gpt-3.5-turbo-0125	5157	95.58	74.58	97.82	84.64	4.74	83.14
5	gpt-4o-mini	5160	95.29	73.17	98.13	83.83	5.11	82.34

Paper 3

Model	ROC AUC	Precision	Recall	F1-Score	Accuracy
SVM-TF	0.9913	0.9855	0.9049	0.9433	0.9855
SVM-TF-Bi	0.9916	0.9856	0.9129	0.9477	0.9865
SVM-TF-Bi-Ada	0.9809	0.9803	0.7925	0.8764	0.9700
SVM-TF-Bi-XGB	0.9840	0.9512	0.8809	0.9145	0.9779
SVM-TF-Bi-Vote	0.9937	0.9925	0.8888	0.9376	0.9842
SVM-DistilBERT-Vote	0.9995	0.9951	0.9828	0.9878	0.9968

Title and Reference	Dataset	Methods	Accuracy
SMS Spam Classification Using Machine Learning Techniques [54]	UCI Machine Learning Repository	SVM	98.797%
SMS Spam Message Detection using Term Frequency-Inverse Document Frequency and Random Forest Algorithm [55]	UCI Machine Learning Repository	TF-IDF with Random Forest	97.50%
Hybrid SMS Spam Filtering System Using Machine Learning Techniques [56]	UCI Machine Learning Repository	K-Means SVM	98.8%
A Robust System For Message Filtering Using An Ensemble Machine Learning Supervised Approach [57]	6000 Messages Data, 1000 Messages are spam	Voting Classifier	98.93%
Semi-supervised novelty detection with one class SVM for SMS spam detection [58]	747 spam, 4827 non-spam messages	One class SVM	98%
Relevant SMS Spam Feature Selection Using Wrapper Approach and XGBoost Algorithm [59]	UCI Machine Learning Repository	XGBoost Classifier	97.64%
This Research	UCI Machine Learning Repository	SVM DistilBERT with Voting Classifier	99.6%

Erros Consistentes entre Modelos

Evidência de Ruído no Dataset

- Durante a avaliação, 23 mensagens apresentaram erros de classificação em todos os 7 modelos.
- Foram identificadas inconsistências semânticas — mensagens legítimas marcadas como spam, ou spam sutis marcados como ham.
- Uma análise manual sugere que parte das falhas se origina de erros de rotulagem na base da UCI SMS Spam Collection, o que significa que a performance real dos modelos pode ser ainda melhor do que a métrica medida.

```
3174,spam,Sorry I missed your call let's talk when you have the time. I'm on 07090201529,
ham,7,"['gpt-4.1-mini', 'gpt-3.5-turbo-0125', 'gpt-5-nano', 'gpt-5', 'gpt-5-mini',
'gpt-4o-mini', 'gpt-4.1']"
3576,ham,"No dude, its not fake..my frnds got money, thts y i'm reffering u..if u member
wit my mail link, u vl be credited <#> rs and il be getiing <#> rs..i can draw my acc wen
it is <#> rs..",spam,7,"['gpt-4.1-mini', 'gpt-3.5-turbo-0125', 'gpt-5-nano', 'gpt-5',
'gpt-5-mini', 'gpt-4o-mini', 'gpt-4.1']"
3819,spam,TBS/PERSOLVO. been chasing us since Sept for£38 definitely not paying now thanks
to your information. We will ignore them. Kath. Manchester.,ham,7,"['gpt-4.1-mini', 'gpt-3.
5-turbo-0125', 'gpt-5-nano', 'gpt-5', 'gpt-5-mini', 'gpt-4o-mini', 'gpt-4.1']"
3980,ham,"Send ur birthdate with month and year, I will tel u ur LIFE PARTNER'S name. and
the method of calculation. Reply must.",spam,7,"['gpt-4.1-mini', 'gpt-3.5-turbo-0125',
'gpt-5-nano', 'gpt-5', 'gpt-5-mini', 'gpt-4o-mini', 'gpt-4.1']"
4411,ham,I (Career Tel) have added u as a contact on INDYAROCKS.COM to send FREE SMS. To
remove from phonebook - sms NO to <#>,spam,7,"['gpt-4.1-mini', 'gpt-3.5-turbo-0125',
'gpt-5-nano', 'gpt-5', 'gpt-5-mini', 'gpt-4o-mini', 'gpt-4.1']"
4576,spam,"Goal! Arsenal 4 (Henry, 7 v Liverpool 2 Henry scores with a simple shot from 6
yards from a pass by Bergkamp to give Arsenal a 2 goal margin after 78 mins.",ham,7,"
['gpt-4.1-mini', 'gpt-3.5-turbo-0125', 'gpt-5-nano', 'gpt-5', 'gpt-5-mini', 'gpt-4o-mini',
'gpt-4.1']"
4981,spam,dating:i have had two of these. Only started after i sent a text to talk sport
radio last week. Any connection do you think or coincidence?,ham,7,"['gpt-4.1-mini', 'gpt-3.
5-turbo-0125', 'gpt-5-nano', 'gpt-5', 'gpt-5-mini', 'gpt-4o-mini', 'gpt-4.1']"
```

Explicabilidade

Reanálise de Mensagens Ambíguas Explicações com GPT-5-mini

→ Foram selecionadas mensagens classificadas incorretamente por 3 modelos distintos.

→ O objetivo foi pedir ao modelo GPT-5-mini que reclassificasse e justificasse a decisão.

- Cada item do *batch* inclui:
 - O texto original da mensagem.
 - Um novo system prompt explicativo, projetado para gerar raciocínios linguísticos, estruturais e contextuais.

```
SYSTEM_PROMPT = (  
    "You are a binary text classifier for SMS messages.  
    Classify each message as exactly one of the following labels:  
    - ham: legitimate, personal, or non-promotional content  
    - spam: promotional, fraudulent, or unsolicited content  
  
    For each message, decide the correct label ('ham' or 'spam')  
    and explain your reasoning in this format:  
    Label: <ham|spam>  
    Reasoning:  
    1. Linguistic cues: <words, tone, or phrases that influenced the  
    decision>  
    2. Structural cues: <formatting, punctuation, or style patterns>  
    3. Contextual interpretation: <intent or purpose inferred from  
    message>  
    4. Decision summary: <short justification connecting cues to  
    final label>"  
)
```

Explicabilidade

Falso Negativo Ambiguidade

- O texto parece um aviso legítimo, mas mensagens desse tipo eram usadas por spammers para validar números.
- O GPT-5-mini interpretou apenas o tom informativo, ignorando o padrão típico de spam técnico.
- É discutível se o modelo deveria marcar como SPAM, visto que é difícil inferir só pelo texto neste caso.

id: 3945

Label Original: spam

Texto SMS:

"Missed call alert. These numbers called but left no message. 07008009200"

Explicação do erro:

1. Linguistic cues: The message uses neutral, informational language ("Missed call alert", "called but left no message") without persuasive, promotional, or urgent phrases typical of spam (e.g., "win", "free", "act now").
2. Structural cues: Short, factual structure with a single phone number listed; lacks links, offers, strange punctuation or call-to-action that often indicate spam or phishing.
3. Contextual interpretation: The intent appears to be a simple notification about missed calls, a common legitimate service provided by telecom operators or call-tracking services.
4. Decision summary: The message is an informational alert rather than a promotional or fraudulent solicitation, so it is classified as ham.

Explicabilidade

Falso Positivo

Viés de segurança/conteúdo

→ Mensagens sexuais pessoais são rotuladas como ham no dataset original.

→ O modelo interpretou o tom explícito como spam sexual, reproduzindo um viés de segurança/conteúdo.

→ Confusão entre conteúdo explícito e intenção criminosa.

id: 4312

Label Original: ham

Texto SMS:

"Good. do you think you could send me some pix? I would love to see your top and bottom..."

Explicação do erro:

"1. Linguistic cues: Uses requests for ""pix"" and explicit phrasing (""your top and bottom""), sexual solicitation language indicating an attempt to obtain intimate images.

2. Structural cues: Short, direct imperative/question format asking for content, informal shorthand (""pix""), and trailing ellipsis implying persistence or suggestiveness.

3. Contextual interpretation: The intent is to solicit explicit photos—unsolicited sexual content—which fits patterns of spam/unwanted solicitation even if not commercial.

4. Decision summary: The message is an unsolicited sexual request for images (solicitation), so it should be classified as spam."

Dataset

Enron Spam Corpus

- Conjunto público de e-mails rotulados, amplamente utilizado em pesquisas sobre classificação de spam e filtragem de mensagens.
- Contém aproximadamente 33.000 e-mails, divididos entre mensagens legítimas (ham) e mensagens de spam.
- As mensagens foram coletadas dos servidores da empresa Enron Corporation, posteriormente analisadas e rotuladas manualmente para compor o Enron Spam Corpus.

Paper

A pipeline and comparative study of 12 machine learning models for text classification

de Tommaso Occhipinti, Simon Rogers e Andrea Angione
2022

→ Propõe e avalia um pipeline unificado para otimizar o pré-processamento, a seleção de atributos e os hiperparâmetros de 12 algoritmos de aprendizado de máquina aplicados à classificação de textos, com foco na detecção de spam no conjunto Enron.

→ Algoritmos avaliados

1. Naïve Bayes (NB)

- Multinomial Naïve Bayes
- Gaussian Naïve Bayes
- Bernoulli Naïve Bayes

2. Support Vector Machines (SVM)

- Linear SVM
- Polynomial SVM (grau 3)
- Sigmoidal SVM
- Radial Basis Function (RBF) SVM

3. k-Nearest Neighbours (kNN)

- Variantes: brute force, ball tree e KD-tree

4. Multilayer Perceptron Neural Network (MPNN)

- Rede neural com algoritmo LBFGS

5. Logistic Regression (LR)

6. Random Forest (RF)

7. Extreme Gradient Boost (XGBoost)



Benchmark

gtp-4.1-mini

→ Avaliação no **Enron Spam Corpus** com **33 000** e-mails balanceados (ham/spam).

→ Teste replicado sob as mesmas condições do benchmark SMS, via **OpenAI Batch API**.

```
SYSTEM_PROMPT = (  
    "You are a binary text classifier for SMS messages. "  
    "Classify each message as exactly one of the following labels:\n"  
    "- ham: legitimate, personal, or non-promotional content\n"  
    "- spam: promotional, fraudulent, or unsolicited content\n\n"  
    "Respond with only the label – 'ham' or 'spam' – without explanation or punctuation."  
)
```

```
10,ham,"Subject: anyone else read this ?
has anyone , or everyone , read the latest business practice proposal
that was released by nerc on tuesday ? it is esc ' s scheduling and approval
practices submitted by the multiple task groups - scheduling process ,
approval , schedule timing , granularity and ramping task groups .
specifically , they are recommending changing the ramp time to a
10 - minute straddle ramp ( : 55 to : 05 ) . in addition , they ' re recommending
allowing schedule changes on the quarter ( as already exists at : 15 , : 30 ,
: 45 ) with a 10 minute ramp . their reasoning is , and this is a quote , "" large
generators that require longer ramp times ( e . g . 20 or 30 minutes ) , could
partially ramp for 10 minutes across the schedule start , then finish with
another 10 - minute ramp across the next quarter to complete the full ramp .
this would accommodate areas such as florida and wsc , where longer ramp
times are needed . ""
their asking for feedback for the public posting . this proposal has
a serious impact on our recommendations . anyone have any comments ? i ' m
attaching a copy of the document . it ' s 21 pages long , about 15 of them on
frag tags . the ramping stuff and recommendations are the last couple of
pages .
carmine tilghman
tucson electric power
wholesale marketing
( 520 ) 745 - 7108
>
- esc scheduling practices ( multiple task groups ) c advena . doc"
```

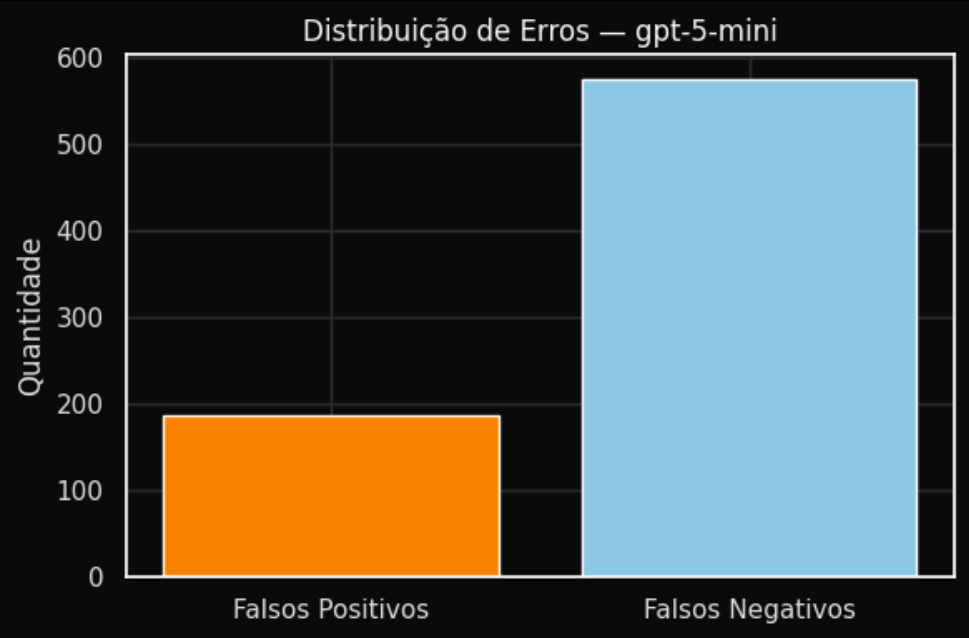
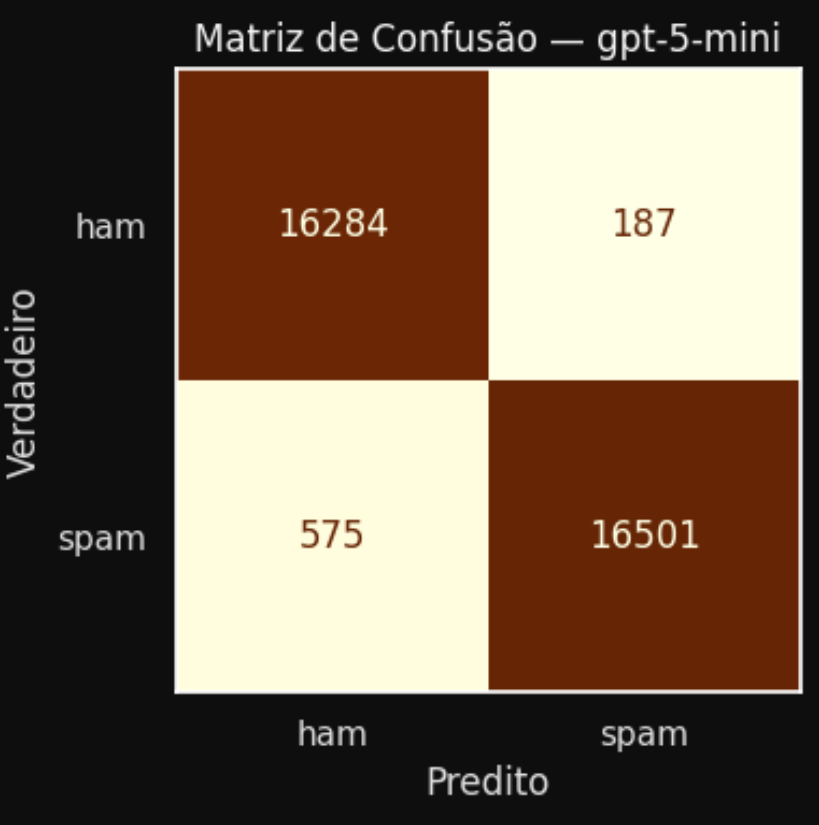
```
17653,spam,"Subject: application confirmation
hello ,
we were reviewing your record and noticed that your mortgage rate was over 6 % .
please fill out the form at this webpage to complete the process :
http : / / www . loant . com / ? partid = rm 2342
we look forward to hearing from you .
regards ,
berrison united , llc .
not interested"
```

```
{"custom_id": "16903", "method": "POST", "url": "/v1/chat/completions", "body": {"model": "gpt-5-mini",
"messages": [{"role": "system", "content": "You are an expert email classifier. Classify each email
message as exactly one of the following labels:\n- ham: normal, work-related, personal, or legitimate
correspondence\n- spam: unsolicited, promotional, fraudulent, or irrelevant content\n\nRespond with only
the label - 'ham' or 'spam' - without explanation or punctuation."}, {"role": "user", "content":
"Subject: don ' t click here\nthousands of people worldwide received their 10 , 000 you can either help
us get you 10 , 000 $ or let us do all the work for you . all you need to know is that by visiting the
site below you will understand how it ' s done ! http : / / www . giftaffiliatesclub . com"}]}}
```

Resultados

Esta Pesquisa

	model	samples	accuracy	precision	recall	f1	bh	mcc
0	gpt-5-mini	33547	97.73	98.88	96.63	97.74	1.14	95.48



Paper 4

Classifier	Time (s)	Precision	Recall	F-score
kNN	0.217	82%	81%	83%
MPNN	0.007	92%	91%	92%
Logistic Regression	0.006	89%	87%	89%
Random Forest	0.025	94%	94%	94%
XGBoost	0.017	94%	93%	94%
Multinomial NB	0.009	85%	85%	85%
Gaussian NB	0.011	87%	85%	87%
Bernoulli NB	0.008	91%	91%	91%
RBF SVM	3.661	77%	58%	71%
Linear SVM	0.212	89%	88%	89%
Poly SVM	0.675	25%	50%	67%
Sigmoid SVM	0.806	76%	56%	70%

Reflexões

Contaminação de Dados × Generalização Real

→ O desempenho excepcional no Enron Spam Corpus ($F1 \approx 97,7\%$) levanta uma dúvida legítima: os modelos GPT podem já ter sido expostos a esse dataset durante o pré-treinamento.

→ Isso implica que parte da performance pode refletir memorização — e não generalização verdadeira.

→ Ao mesmo tempo, essa vantagem não é apenas “sorte”:

- os LLMs têm capacidade linguística real, entendendo contexto, ironia, estrutura retórica e o tom de mensagens — aspectos que modelos clássicos de ML não captam.

→ O desafio, portanto, é distinguir entre conhecimento legítimo e lembrança do treino.

→ Próximos passos:

- Avaliar em datasets inéditos ou sintéticos, criados após o corte de treinamento dos modelos.
- Realizar testes cruzados com e-mails modernos, SMS e redes sociais.
- Explorar robustez semântica, perturbando nomes, domínios e datas.
- Investigar até onde o modelo compreende a estrutura do spam versus exemplos específicos já vistos.

